Table 1: Raw win rate by number of Tanks fielded

Tanks	Teams	Win rate (%)	vs. two tanks
0	1,990	18.49	-33.81
1	89,534	39.44	-12.86
2	**533,702**	**52.30**	—
3	24,456	41.28	-11.02
4	288	34.72	-17.58

Composition from each player's play-time-dominant hero. The relationship is an inverted U peaking at the 2–2–2 standard that 76% of teams field. A term linear in tank counts cannot represent this shape; because teams fielding too few Tanks outnumber those fielding too many by roughly four to one, a fitted slope is dragged positive. This is the artifact the within-role constraint removes.

Hero Adjusted Plus–Minus in *Marvel Rivals*

Intention-to-treat estimates from 477,483 competitive matches

Season 19, PC ranked. Prepared 8 September 2026. Commit `31cfdff`.

Specification. One observation per match; the outcome is victory for side 0. Each hero enters as a signed contrast (own-side weight less opposing-side weight) under *starting-lineup* attribution (W0), reduced onto a basis that sums to zero *within each role*, so every hero receives both a coefficient and a standard error. Controls: a separate intercept per map, the pre-match rank-score differential, composition-shape indicators, and 100 team-up contrasts. Estimated by unpenalised logit with heteroskedasticity-robust (HC1) standard errors.

The estimand is intention-to-treat. W0 is the hero each player *started* on, recovered from entry 1 of the API hero array, which is chronological by first appearance. It is the only attribution fixed before any outcome information exists. A player who starts on a hero and abandons it after thirty seconds still carries full weight, so these effects are diluted by non-compliance — but how often a hero gets abandoned is part of its value, not an error to remove.

Why not play-time weighting. Appendix Table A1 shows effect magnitude rising monotonically with how much post-outcome information the attribution uses. The best-*fitting* rule is the most contaminated: play-time weighting predicts better precisely because its regressors encode what happened.

Why estimates are within role. Cross-role comparisons are not a hero property but a composition property. The composition-shape block can reproduce the role-count differential *exactly*, so under a single global sum-to-zero constraint the hero and shape blocks are collinear in that direction and are separated only by the 0.16% of teams pooled into an "other" category. Constraining β to sum to zero within each role makes it orthogonal to every role indicator, so composition can only land in the shape block, where the full sample identifies it. Refitting under this constraint leaves the estimates numerically unchanged (Spearman 0.9999, mean absolute difference 0.010 pp), confirming that the earlier post-hoc projection was already removing exactly the ill-conditioned direction.

Table 2: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	-1.1×10^{-16}	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.949$	up from 0.703 before correction
Within-role reparameterisation	$\rho = 0.9999$	estimates numerically unchanged
Camp indexing	49.87% camp 0	absolute map side, no crawl selection

The placebo shuffles hero assignment against outcomes and refits; a placebo that failed would invalidate the specification. It was run under play-time attribution on the same pipeline. Camp-0 win rate varies 49.42%–52.52% across the 16 maps, which is why each map carries its own intercept.

Table 3: Hero adjusted plus–minus, relative to an average hero of the same role

	APM	Std. error	95% CI	q
Panel A: Tank (15 heroes)				
The Thing	5.46***	(0.204)	[5.06, 5.86]	0.000
Magneto	1.96***	(0.139)	[1.69, 2.24]	0.000
Angela	1.87***	(0.261)	[1.36, 2.38]	0.000
Peni Parker	1.51***	(0.328)	[0.86, 2.15]	0.000
Hulk	1.33***	(0.219)	[0.90, 1.76]	0.000
Thor	1.29***	(0.219)	[0.86, 1.72]	0.000
The Hood	0.45**	(0.172)	[0.11, 0.78]	0.012
Devil Dinosaur	-0.03	(0.231)	[-0.48, 0.42]	0.907
Groot	-0.67**	(0.290)	[-1.23, -0.10]	0.027
Venom	-0.93***	(0.219)	[-1.36, -0.51]	0.000
Doctor Strange	-0.98***	(0.202)	[-1.37, -0.58]	0.000
Captain America	-1.22***	(0.352)	[-1.91, -0.53]	0.001
Emma Frost	-1.77***	(0.192)	[-2.15, -1.39]	0.000
Rogue	-3.41***	(0.344)	[-4.09, -2.74]	0.000
Deadpool (Vanguard)	-4.86***	(0.252)	[-5.36, -4.37]	0.000
Panel B: Damage (27 heroes)				
Black Cat	4.85***	(0.260)	[4.34, 5.36]	0.000
Magik	3.47***	(0.219)	[3.04, 3.90]	0.000
Storm	3.09***	(0.305)	[2.49, 3.68]	0.000
Daredevil	2.92***	(0.217)	[2.50, 3.35]	0.000
Mister Fantastic	2.52***	(0.351)	[1.83, 3.21]	0.000
Iron Man	1.34***	(0.279)	[0.79, 1.89]	0.000
Iron Fist	1.32***	(0.224)	[0.88, 1.76]	0.000
Hela	1.14***	(0.199)	[0.75, 1.53]	0.000
Winter Soldier	1.00***	(0.141)	[0.73, 1.28]	0.000
Wolverine	0.82**	(0.317)	[0.20, 1.44]	0.012
Spider-man	0.58***	(0.175)	[0.24, 0.93]	0.001
Moon Knight	0.10	(0.297)	[-0.49, 0.68]	0.792
Black Widow	0.02	(0.200)	[-0.37, 0.42]	0.907
Blade	-0.20	(0.317)	[-0.82, 0.43]	0.578
Black Panther	-0.21	(0.239)	[-0.68, 0.26]	0.423
Human Torch	-0.28	(0.351)	[-0.97, 0.41]	0.468
Elsa Bloodstone	-0.39	(0.229)	[-0.83, 0.06]	0.108
Psylocke	-0.58	(0.342)	[-1.25, 0.09]	0.106
Deadpool (Duelist)	-0.67***	(0.220)	[-1.10, -0.24]	0.003
Namor	-0.82***	(0.243)	[-1.30, -0.34]	0.001

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Table 3, continued

	APM	Std. error	95% CI	q
Hawkeye	-1.22***	(0.341)	[-1.89, -0.55]	0.001
The Punisher	-1.81***	(0.262)	[-2.33, -1.30]	0.000
Star-lord	-2.41***	(0.300)	[-3.00, -1.82]	0.000
Scarlet Witch	-2.49***	(0.414)	[-3.30, -1.68]	0.000
Squirrel Girl	-3.78***	(0.350)	[-4.46, -3.09]	0.000
Cyclops	-4.04***	(0.195)	[-4.42, -3.66]	0.000
Phoenix	-4.28***	(0.345)	[-4.96, -3.61]	0.000
Panel C: Support (13 heroes)				
Mantis	6.76***	(0.181)	[6.40, 7.11]	0.000
Rocket Raccoon	4.14***	(0.150)	[3.85, 4.44]	0.000
Ultron	3.59***	(0.297)	[3.01, 4.17]	0.000
Adam Warlock	2.08***	(0.315)	[1.47, 2.70]	0.000
Loki	1.18***	(0.268)	[0.65, 1.70]	0.000
Cloak & Dagger	-0.04	(0.170)	[-0.37, 0.29]	0.852
Gambit	-0.30	(0.210)	[-0.71, 0.11]	0.178
Invisible Woman	-1.38***	(0.151)	[-1.67, -1.08]	0.000
Jeff The Land Shark	-1.75***	(0.233)	[-2.21, -1.29]	0.000
Jubilee	-1.91***	(0.172)	[-2.25, -1.58]	0.000
Luna Snow	-3.01***	(0.167)	[-3.33, -2.68]	0.000
White Fox	-3.48***	(0.208)	[-3.89, -3.07]	0.000
Deadpool (Strategist)	-5.89***	(0.210)	[-6.30, -5.48]	0.000
Matches	477,483			
Player–matches	5,729,796			
Heroes	55			
Log-likelihood	-320,067.7			

Notes. All 55 heroes are listed. Units are percentage points of win probability at a balanced match ($\partial p/\partial x = \beta/4$): the change from fielding this hero at match start in place of an average hero of the same role. Standard errors in parentheses are HC1. Stars and q denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: $^{*}q < 0.10$, $^{**}q < 0.05$, $^{***}q < 0.01$; 10 of the 55 are not distinguishable from their role average. Sample: 482,997 matches crawled, less 2,139 containing draws (`is_win` = 2), 19 with incomplete play time, and 3,356 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped. Appendix Table A1 repeats every hero under the two contaminated attribution rules.

Appendix

Table A1: Every hero under all three attribution rules, same sample

	W0 starting	W2 dominant	W1 play-time	W1−W0
Panel A: Tank				
The Thing	5.46	4.94	7.21	+1.75
Magneto	1.96	0.55	3.57	+1.60
Angela	1.87	2.00	2.68	+0.81
Peni Parker	1.51	10.18	7.41	+5.90
Hulk	1.33	2.03	3.55	+2.21
Thor	1.29	0.73	0.65	-0.64
The Hood	0.45	0.35	6.52	+6.08
Devil Dinosaur	-0.03	-0.56	-2.24	-2.21
Groot	-0.67	-1.52	-3.79	-3.13
Venom	-0.93	-1.25	-1.35	-0.41
Doctor Strange	-0.98	-2.55	-6.82	-5.85
Captain America	-1.22	-0.51	-1.78	-0.57
Emma Frost	-1.77	-4.62	-6.68	-4.91
Rogue	-3.41	-3.98	-1.78	+1.64
Deadpool (Vanguard)	-4.86	-5.79	-7.15	-2.28
Panel B: Damage				
Black Cat	4.85	7.74	11.62	+6.77
Magik	3.47	7.94	14.57	+11.11
Storm	3.09	6.57	10.27	+7.18
Daredevil	2.92	5.70	8.47	+5.55
Mister Fantastic	2.52	3.02	3.95	+1.43
Iron Man	1.34	2.21	3.10	+1.76
Iron Fist	1.32	2.84	4.52	+3.19
Hela	1.14	1.04	1.22	+0.08
Winter Soldier	1.00	-0.07	0.17	-0.83
Wolverine	0.82	0.62	0.21	-0.61
Spider-man	0.58	1.48	2.90	+2.32
Moon Knight	0.10	-2.01	-3.37	-3.47
Black Widow	0.02	-0.96	-1.24	-1.26
Blade	-0.20	0.28	1.09	+1.28
Black Panther	-0.21	1.20	2.87	+3.08
Human Torch	-0.28	1.74	4.56	+4.84
Elsa Bloodstone	-0.39	-0.46	-0.12	+0.26

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Table A1, continued

	W0	W2	W1	W1−W0
Psylocke	-0.58	0.08	1.54	+2.12
Deadpool (Duelist)	-0.67	0.48	1.85	+2.52
Namor	-0.82	-3.87	-7.57	-6.75
Hawkeye	-1.22	-1.31	-2.25	-1.03
The Punisher	-1.81	-5.38	-10.45	-8.64
Star-lord	-2.41	-2.64	-4.05	-1.64
Scarlet Witch	-2.49	-4.65	-5.91	-3.43
Squirrel Girl	-3.78	-8.62	-16.61	-12.83
Cyclops	-4.04	-5.45	-7.65	-3.61
Phoenix	-4.28	-7.54	-13.67	-9.38
Panel C: Support				
Mantis	6.76	9.88	13.50	+6.74
Rocket Raccoon	4.14	4.50	5.18	+1.03
Ultron	3.59	6.92	6.76	+3.17
Adam Warlock	2.08	4.11	3.93	+1.84
Loki	1.18	2.14	1.84	+0.66
Cloak & Dagger	-0.04	-1.84	0.38	+0.42
Gambit	-0.30	-2.07	-0.64	-0.34
Invisible Woman	-1.38	-3.32	-5.39	-4.01
Jeff The Land Shark	-1.75	-3.04	-5.01	-3.27
Jubilee	-1.91	-1.94	0.32	+2.23
Luna Snow	-3.01	-5.00	-7.25	-4.24
White Fox	-3.48	-4.73	-6.69	-3.21
Deadpool (Strategist)	-5.89	-5.62	-6.91	-1.02
Log-likelihood	−320,068	−305,869	−297,648	
Effect s.d.	2.60	4.27	6.38	
ρ with W0	—	0.937	0.891	

Notes. Within-role estimates in percentage points, all three fitted on the same 477,483 matches. Effect magnitude rises monotonically with how much post-outcome information the rule uses, while the ranking is broadly preserved. W1−W0 is reported as a difference rather than a ratio because a ratio diverges for heroes whose W0 estimate is near zero. Higher log-likelihood here indicates worse identification, not a better model: play-time weighting predicts better precisely because its regressors encode what happened.

Table A2: Team-up synergies, all 100 pairs

Team-up	Premium	Std. err.	Team-up	Premium	Std. err.
Squirrel Girl × Iron Man	8.66	(0.965)	Angela × Loki	1.35	(0.957)
Blade × Moon Knight	8.30	(0.851)	Invisible Woman × Human Torch	1.32	(0.507)
Rogue × Gambit	6.95	(0.472)	Gambit × Magneto	1.31	(0.277)
Adam Warlock × Ultron	6.82	(0.768)	Star-lord × Groot	1.24	(0.784)
Iron Fist × White Fox	6.61	(0.498)	Elsa Bloodstone × Devil Dinosaur	1.21	(0.583)
Mantis × Star-lord	4.27	(0.503)	The Thing × Invisible Woman	1.20	(0.358)
The Punisher × Rocket Raccoon	4.13	(0.422)	Rogue × Magneto	1.07	(0.501)
Storm × Jeff The Land Shark	3.86	(0.667)	Cloak & Dagger × Luna Snow	1.07	(0.318)
Human Torch × Jubilee	3.71	(0.469)	Captain America × Winter Soldier	0.98	(0.672)
Loki × Mantis	3.53	(0.667)	Namor × Hulk	0.86	(0.585)
Doctor Strange × Hulk	3.47	(0.603)	Devil Dinosaur × Jeff The Land Shark	0.83	(0.679)
Magik × The Hood	3.36	(0.295)	Phoenix × Rogue	0.79	(1.066)
Adam Warlock × Storm	3.28	(1.149)	Iron Fist × The Thing	0.73	(0.490)
Luna Snow × White Fox	3.19	(0.447)	Emma Frost × Luna Snow	0.70	(0.332)
Peni Parker × Rocket Raccoon	3.05	(0.543)	Wolverine × Cyclops	0.68	(0.881)
Cyclops × Phoenix	2.79	(0.641)	White Fox × Psylocke	0.61	(0.778)
Mantis × Adam Warlock	2.76	(0.831)	Peni Parker × Black Panther	0.52	(1.041)
Cloak & Dagger × The Hood	2.60	(0.233)	Rocket Raccoon × Squirrel Girl	0.50	(0.726)
Doctor Strange × Invisible Woman	2.58	(0.326)	Loki × Hela	0.45	(0.708)
Scarlet Witch × Jubilee	2.47	(0.510)	Hela × Namor	0.21	(0.552)
Jubilee × The Hood	2.46	(0.236)	Gambit × Jubilee	0.21	(0.322)
Black Widow × Phoenix	2.44	(0.859)	Mister Fantastic × The Thing	0.17	(0.773)
Hulk × Captain America	2.39	(0.998)	Luna Snow × Adam Warlock	0.16	(0.763)
Hawkeye × Psylocke	2.35	(0.907)	Devil Dinosaur × The Punisher	0.10	(0.700)
Squirrel Girl × Spider-man	2.17	(0.959)	Namor × Luna Snow	0.09	(0.483)
Ultron × Peni Parker	2.11	(1.072)	Spider-man × Venom	0.06	(0.424)
Spider-man × Peni Parker	2.02	(0.714)	Groot × Jeff The Land Shark	-0.02	(0.819)
Winter Soldier × Elsa Bloodstone	2.00	(0.442)	Psylocke × Cloak & Dagger	-0.03	(0.514)
Phoenix × Hela	1.98	(0.808)	Venom × Blade	-0.04	(0.750)
Moon Knight × Cloak & Dagger	1.97	(0.465)	Captain America × Thor	-0.06	(1.079)
Star-lord × Adam Warlock	1.95	(1.190)	Moon Knight × Elsa Bloodstone	-0.06	(0.940)
Iron Man × Thor	1.87	(0.688)	Winter Soldier × The Punisher	-0.07	(0.538)
Scarlet Witch × Doctor Strange	1.85	(0.628)	Magneto × Emma Frost	-0.12	(0.313)
Groot × Mantis	1.84	(0.535)	Black Cat × Black Panther	-0.25	(1.018)
White Fox × Black Cat	1.77	(0.587)	Jubilee × Blade	-0.41	(0.519)
Black Panther × Storm	1.77	(0.996)	Invisible Woman × Mister Fantastic	-0.44	(0.614)
Hela × Venom	1.74	(0.552)	The Thing × Human Torch	-0.45	(0.548)

Table A2, continued

Team-up	Premium	Std. err.	Team-up	Premium	Std. err.
Thor × Hela	1.71	(0.552)	Psylocke × Emma Frost	-0.49	(0.615)
Venom × Phoenix	1.67	(0.596)	Magneto × Scarlet Witch	-0.55	(0.511)
Mister Fantastic × Rocket Raccoon	1.62	(0.591)	Rocket Raccoon × Groot	-0.55	(0.518)
Blade × Captain America	1.62	(0.981)	Thor × Angela	-0.81	(1.007)
The Punisher × Daredevil	1.58	(0.850)	Iron Man × Hulk	-0.86	(0.796)
Hawkeye × Cloak & Dagger	1.58	(0.542)	Cyclops × Gambit	-1.00	(0.411)
Ultron × Iron Man	1.54	(0.864)	Storm × Thor	-1.17	(0.733)
Emma Frost × Mantis	1.50	(0.363)	Hulk × Wolverine	-1.39	(0.805)
Daredevil × Black Widow	1.48	(0.728)	Black Widow × Hawkeye	-1.54	(1.126)
Wolverine × Gambit	1.47	(0.581)	Daredevil × Iron Fist	-1.80	(0.741)
Magik × Doctor Strange	1.45	(0.389)	Human Torch × Storm	-1.80	(1.057)
Jeff The Land Shark × Venom	1.41	(0.600)	Black Panther × Magik	-2.09	(0.666)
Angela × Star-lord	1.39	(1.006)	Black Cat × Spider-man	-2.24	(0.627)

Notes. All 100 team-ups are listed. Read down the left column, then the right. Each is labelled by its member heroes, anchor first. The premium is what a pair earns *beyond* what its two heroes contribute individually, in percentage points, under W0 attribution. Four further team-ups are absent because they are defined against hero id 1057, base “Deadpool”, whose plays are all recorded under his three role variants, leaving those columns structurally empty.

Limitations.

1. *Magnitudes depend on the attribution rule; the ranking is far more stable.* Across W0, W2 and W1 on an identical sample the within-role standard deviation runs 2.60, 4.27 and 6.38 points while rank agreement with W0 holds at $\rho = 0.937$ and 0.891. W0 is reported because it is the only rule fixed before any outcome information exists, but it is intention-to-treat and diluted by mid-match abandonment. Read the ordering as the firmer result.

2. *These are adjusted associations, not causal effects.* Hero choice is not random. Pre-match rank score controls for *general* player skill but not for skill *on that specific hero*, and a one-trick is not a random player assigned that hero. Per-player hero playtime would measure this confounder directly; it was costed and declined, so it is bounded rather than eliminated.

3. *Intervals are analytic, not bootstrapped.* Matches are not independent draws — each contains twelve players and so belongs to twelve overlapping player clusters. The specified cluster bootstrap would need roughly 656 hours at this scale under the current implementation, so HC1 intervals are reported instead. They ignore cross-classified dependence and are therefore *too narrow*; point estimates are unaffected.

4. *Composition control is coarse in the tail.* Twelve shapes carry their own indicator and the remainder share one, covering 0.16% of team-instances whose win rates span 0.0–53.8%.

5. *The sample is rank-selected.* Profile privacy rises from 0% below 4,000 rank score to 70% above 5,000, and the crawl can only expand through public players, so high-elo matches are structurally under-sampled.